

Capacity Planning Example: URL Shortener

Worked Example with All 10 Steps + Request Size Calculation

Assumptions:

- Total Users: 100M
- DAU: 10M (10% engagement)
- Actions per User per Day: Redirects: 100, Shortens: 0.01 (~100K/day)
- Request size: Redirect log calculated below

Step 1. Requests per Second (RPS)

Formula: $RPS = (DAU \times \text{Actions per User per Day}) / 86,400$

Redirects = $(10M \times 100) / 86,400 \approx 11,574$ RPS

Shortens = $100K / 86,400 \approx 1.15$ RPS

Step 2. Average Request Size + Throughput

How to calculate Avg Request Size:

shortCode = 10 bytes

timestamp = 8 bytes

IP address = 16 bytes

user agent = 100 bytes

geo = 10 bytes

Total ≈ 144 bytes $\rightarrow$ round to ~ 200 bytes with overhead

Formula: $\text{Throughput (bytes/sec)} = RPS \times \text{Avg Request Size}$

Redirect logs = $11,574 \times 200B \approx 2.3$ MB/sec

Step 3. Daily Data Growth

Formula: $\text{Daily Growth} = \text{Throughput} \times 86,400$

= $2.3 \text{ MB/sec} \times 86,400 \approx 199$ GB/day

Step 4. Annual Data Growth

Formula: $\text{Annual Growth} = \text{Daily Growth} \times 365$

= $199 \text{ GB/day} \times 365 \approx 73$ TB/year

Step 5. Node Capacity Estimation

Formula: $\text{Nodes} = \text{Required RPS} \div \text{RPS per Node}$

Suppose 1 DB node can handle 2K writes/sec.

Required = $11,574 \div 2,000 \approx 6$ nodes (plus replicas).

Step 6. Peak Traffic Planning

Formula: $\text{Peak RPS} \approx 2-3 \times \text{Avg RPS}$

Peak RPS $\approx 3 \times 11,574 \approx 34,700$ RPS

Step 7. Cache Offload

Formula: Effective DB RPS = Total RPS × (1 – Cache Hit Ratio)

With 95% cache hit ratio: $11,574 \times 0.05 \approx 579$ RPS

Step 8. Network Bandwidth

Formula: Bandwidth (MB/sec) = Throughput ÷ 1,048,576

= $2.3 \text{ MB/sec} \div 1,048,576 \approx 0.0022 \text{ GB/sec}$ (~18.9 GB/day traffic)

Step 9. Latency Budget

Define acceptable latency per request (e.g., <100ms redirect).

Example budget:

- Cache lookup: ~5ms
- Network: ~20ms
- DB fallback: ~50ms
- Buffer: 25ms

Step 10. Cost Estimation

Formula: Cost ≈ (Nodes × Cost per Node) + (Storage × \$/GB) + Network

Example:

- 6 DB nodes × \$500/month = \$3,000/month
- Storage 73 TB/year × \$20/TB = \$1,460/year
- Network ~20GB/day ≈ 600GB/month ≈ \$50/month

Total ≈ \$3,500/month for infra.

Conclusion:

- Reads dominate massively (redirects).
- Core DB load reduced by cache (95% offload).
- Storage growth is in logs (~73TB/year).
- Cost driven by DB nodes and long-term storage.